

Q15

Option A: $R_{tot} = R + \frac{(2R)(R)}{2R+R} = \frac{5R}{3}$.

Power is $P_A = \frac{3V^2}{5R}$

Option B: $R_{tot} = \frac{(2R)R}{2R+R} = \frac{2R}{3}$.

Power is $P_B = \frac{3V^2}{2R}$

Option C: $R_{tot} = 2R$. Power is $P_C = \frac{V^2}{2R}$

Option D: $R_{tot} = R + \frac{R}{2} = \frac{3R}{2}$.

Power is $P_D = \frac{2V^2}{3R}$

Hence the battery in circuit B produces the most power.

Ans: B

